■ Basic Machine Simulator – Grader Guide

1■■ Submission Folder Structure

```
Submission/
  ■ Basic_Machine.jar
  ■ tests/
    MemoryTest.java
    CPUInstructionTest.java
    IntegrationTest.java
    SimulatorRunTest.java
    TestRunner.java
  ■■ run_all.sh
  ■ README_for_Grader.pdf
```

2■■ Requirements

• Java JDK 17 or newer (OpenJDK 21 recommended)
• JavaFX SDK 25 (path example: */Users/<username>/Downloads/javafx-sdk-25/lib*)
• Terminal or Bash shell support (macOS, Linux, or Windows WSL)

3■■ Running the Project

From inside the **Submission** folder, simply run:

```
bash run_all.sh
```

This script will automatically compile, run all tests, and launch the simulator GUI if all tests pass.

4■■ Script Workflow

Step 1: Compile source and test files
Step 2: Run all automated tests
Step 3: Launch the GUI simulator if all tests pass

5■■ Inside the Simulator

Supported Instructions: LDR, LDA, LDX, STR, HLT

GUI Buttons:
• **Load/Load+:** Write octal value into memory @ MAR
• **Store/Store+:** Read from memory @ MAR
• **Run:** Executes until HALT
• **Step:** Executes one cycle
• **IPL:** Load program from load.ld
• **Halt:** Stop execution manually
• **Console Send:** Output message to printer

6■■ Expected Grader Outcome

Upon executing **bash run_all.sh**:
1. The automated tests will run and display results (PASS/FAIL).
2. If all tests pass, the **Basic Machine Simulator GUI** will automatically launch.
3. The grader can interact with the simulator to manually verify program behavior (Load, Run, Step, Halt).

If tests fail, the script will display which specific test failed before launching is disabled.